

Evaluation Framework for Multimodal Language Models Applied to Geospatial Open Data in Agriculture

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Abstract

Abstract example for the Results in Physics Journal. Here you provide a brief summary of the research and the results.

Keywords: keyword 1, keyword 2, keyword 3, keyword 4

1. Introduction

2. Related works

3. Materials and methods

3.1. Evaluation

Comentario

Unlike classical Machine Learning problems, such as classification or regression, evaluating generative artificial intelligence models poses a genuine challenge. This is due to the infinite space of correct *outputs* (responses). The approach in ? was to use the *LLM-as-a-judge* method, a very common solution in the state of the art given its simplicity: it consists of having another language model evaluate the responses. However, this type of evaluation is highly vulnerable to technical hallucinations, suffers from self-reference bias, and tends to lose consistency.

Therefore, **an experimental evaluation methodology has been designed**, which is detailed in this section. This approach, designed to optimize the consumption of operational resources while maintaining statistical robustness, consists of two main stages: evaluation of the complete system and analysis of marginal contributions through an ablation study.

The main challenge of this experimental methodology, and what differentiates it from other existing approaches, is that it must be able to address the evaluation of geospatial context usage in response generation, not merely the quality of the responses themselves. A response that is correct but generic, or even seemingly specific but generated from the model's base knowledge, is not equivalent to a *personalized* response adapted to terrain characteristics. Therefore, multiple stages are combined, different configurations are evaluated, and specific metrics are proposed.

3.2. Parcel Selection and Heterogeneity

A total of 15 parcels were selected, categorized into four different Land Use (LU) types. The distribution of evaluations is

weighted according to the operational utility of the tool for each functional land use, as shown in 1.

To ensure an unbiased evaluation, parcels were selected to represent a high degree of heterogeneity in their physical characteristics, including variable slopes (0.5% to 21.6%), altitudes, irregular shapes, and diversity in vegetation health (NDVI ranges). The parcels are shown in 3.

Table 1: Distribution of parcels by Land Use (LU) type

Land Use	Number of queries
Fruit Trees (FR)	7
Pasture (PA)	5
Unproductive (UN)	2
Forest (FO)	1

3.3. Phase I: Broad Performance Assessment

This phase evaluates the overall system capability across diverse scenarios. A total of 71 queries were designed, distributed as follows:

- **Productive Parcels (FR and PA):** 5 queries per parcel, focused on irrigation, fertilization, pest management, and investment ROI.
- **Non-Productive Parcels (UN and FO):** 4 queries per parcel, consisting of 2 functional queries and 2 *Negative Testing Queries* (scenarios where the system must identify unsuitability).

3.4. Phase II: Ablation Study (Architecture Validation)

To quantify the impact and marginal contribution of each architectural module, an ablation study was conducted using a **factorial design**. Ablation studies consist of the individual analysis of different subsets (configurations) of a system, obtained through systematic activation and deactivation of its components. In this case, with three modules (Orthophoto,

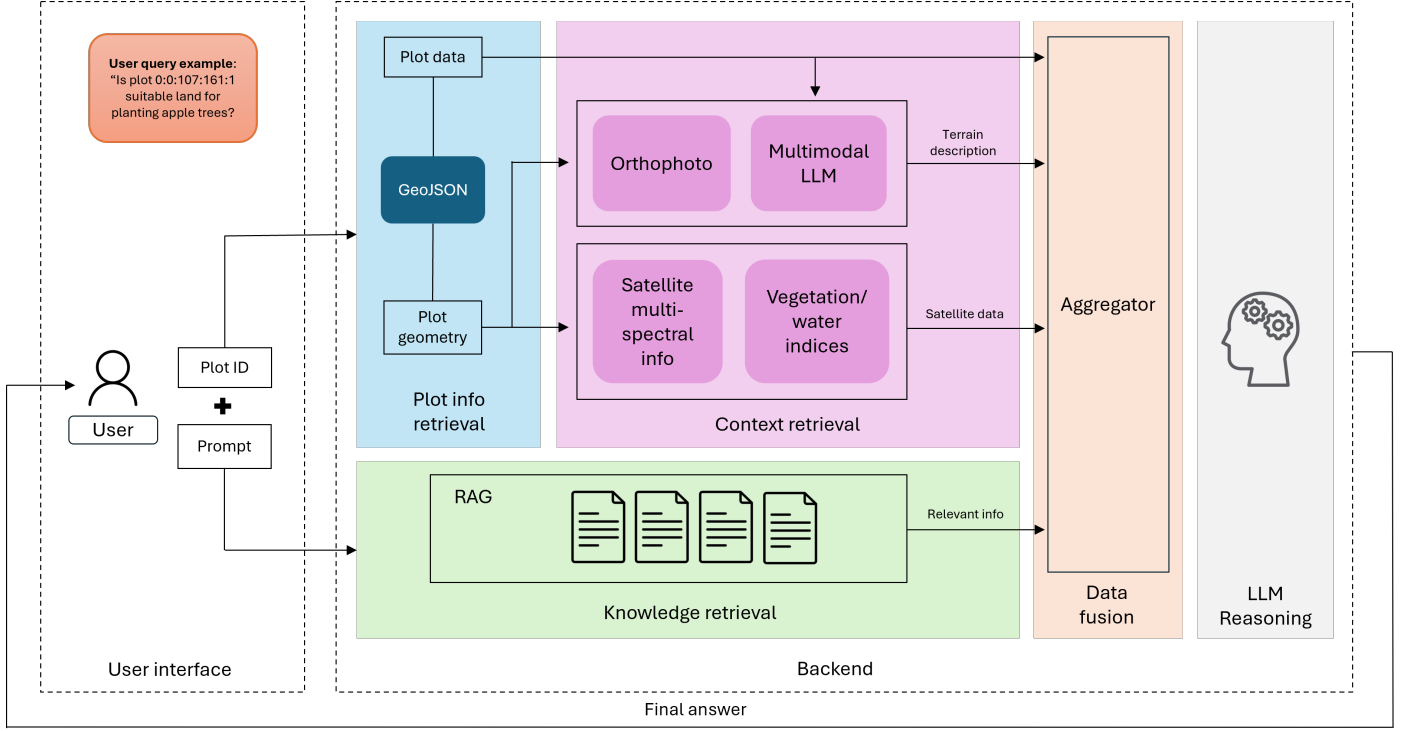


Figure 1:

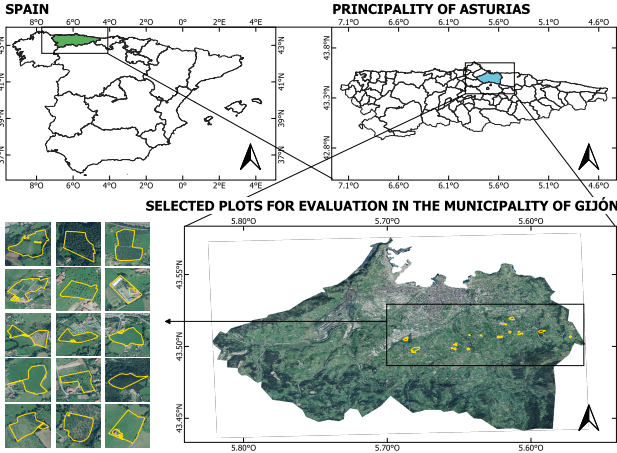


Figure 2:

Satellite, and RAG, previously shown in ??), the factorial design of experiments yields $2^3 = 8$ possible configurations.

- **Configurations:** 8 modular combinations + 1 Zero-Shot baseline (the LLM without parcel context) = 9 total configurations, listed in 2.
- **Sample subset:** 1 representative parcel per Land Use category.
- **Query volume:** 2 specific queries per parcel across the 9 configurations = 72queries.

3.5. Evaluation Volume Summary

The combined results provide a total set of 143 evaluations (3). In Appendix A, a link to the dataset created for this evaluation is included, where all queries, responses, and context can be consulted.

3.6. Evaluation Subjects and Bias Mitigation

To ensure balanced validation, mitigate individual biases, and capture a wide range of perspectives, the evaluation process is carried out independently by **four differentiated evaluators**:

- **Human Evaluators (2):**

- *Expert Evaluator:* PhD in Biology.
- *Control Evaluator:* Non-specialist profile.

- **LLM-as-a-judge (2):**

- *Proprietary Model (Gemini):* Gemini3.1 Flash Lite
- *Open Source Model (Qwen):* Qwen3.5-397B-A17B-Instruct

3.7. Multidimensional Metrics Framework

Response quality is measured according to a multidimensional scale consisting of five metrics, divided into those scored considering context and those that are not, as shown in 4. Scores are given on a *Likert* scale (integer values between 1 and 5).

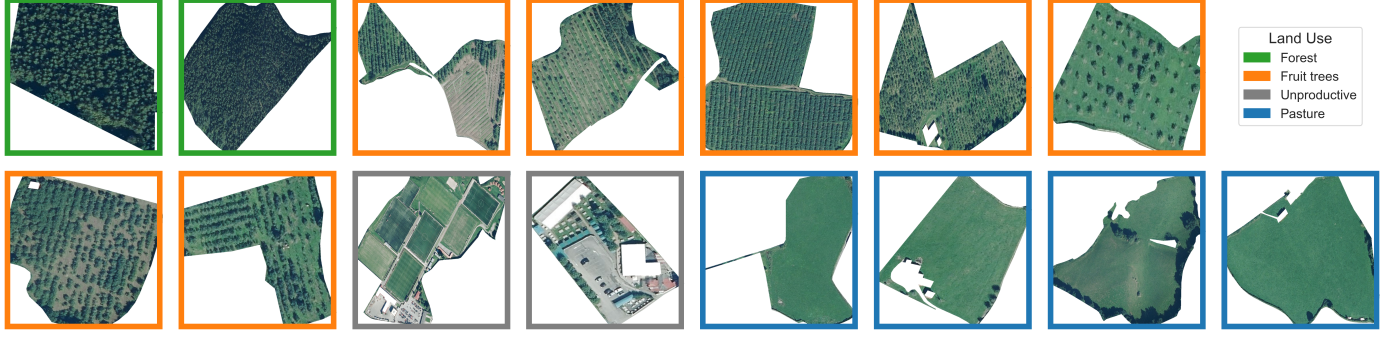


Figure 3: Selection of parcels for tool evaluation, grouped by Land Use. The number of parcels per LU depends on their functional utility in agriculture.

Table 2: Configuraciones modulares del estudio de ablación. PD means Plot Data. Active tool is always enabled because it is responsible for identifying parcel IDs as so.

Modules	RAG	ORTHO	SAT	PD	Active Tool	Code
Baseline	0	0	0	1	1	0
B + SAT	0	0	1	1	1	1
B + ORTHO	0	1	0	1	1	2
B + ORTHO + SAT	0	1	1	1	1	3
B + RAG	1	0	0	1	1	4
B + RAG + SAT	1	0	1	1	1	5
B + RAG + ORTHO	1	1	0	1	1	6
Complete	1	1	1	1	1	7
ZERO SHOT	0	0	0	0	1	ZS

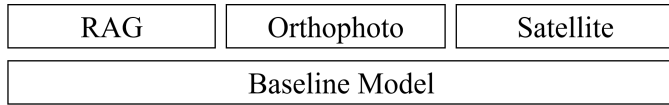


Figure 4:

Table 3: Summary of total query volume in the evaluation.

Evaluation Phase	Methodology	Num. of queries
Broad Assessment	Sequential	71
Ablation Study	Simultaneous	72
Total		143

Table 4: Multidimensional evaluation metrics framework. Note: The *Grounding* metric measures the degree to which responses are derived from the obtained context, rather than generic knowledge, making it the only metric requiring context for evaluation. The *Correctness* metric does not apply to the human control evaluator.

Metric	Type
Correctness	Without Context
Completeness	Without Context
Relevance	Without Context
Utility	Without Context
Grounding	With Context

3.8. Preliminary Results and Discussion

This section presents some preliminary evaluation results. It should be noted that the graphs included here are only an initial overview that sheds light on the system evaluation; more

advanced statistical analyses are reserved for a future scientific publication. First, 5 contains the average scores of each of the 4 evaluators across the 5 metrics, grouped by configuration. From this graph, numerous observations can be made, with the most notable being those related to evaluator correlation, and especially regarding the marginal contributions of the modules. Recalling 2, we see that configurations 2 and 3, as well as 6 and 7, include the orthophoto module; and in all metrics, a higher average score is observed in these cases, allowing us to deduce that the marginal contribution of the orthophoto module is significant.

Similarly, the trend across all metrics is ascending along the x-axis (values are lower in left-side configurations), showing how, for example, system utility in the *Zero Shot* configuration is below 1/5 for most evaluators. Since this configuration corresponds to a standard LLM model without context augmentation, and with configurations progressively ordered by active modules, it can be concluded that **responses, as expected, improve as more geospatial terrain information is added**. This result consequently verifies the initial hypothesis and validates the proposed architecture.

More generally, 6 represents the average metric values from the *Broad* stage evaluations, i.e., the complete system. For all evaluators, the resulting metrics average values very close to 5/5, implying that the system has responded with a high degree of success to the evaluated queries. These preliminary results serve as evidence that the proposed system is capable of generating high-quality, complete, useful, and generally successful responses in the broad sense.

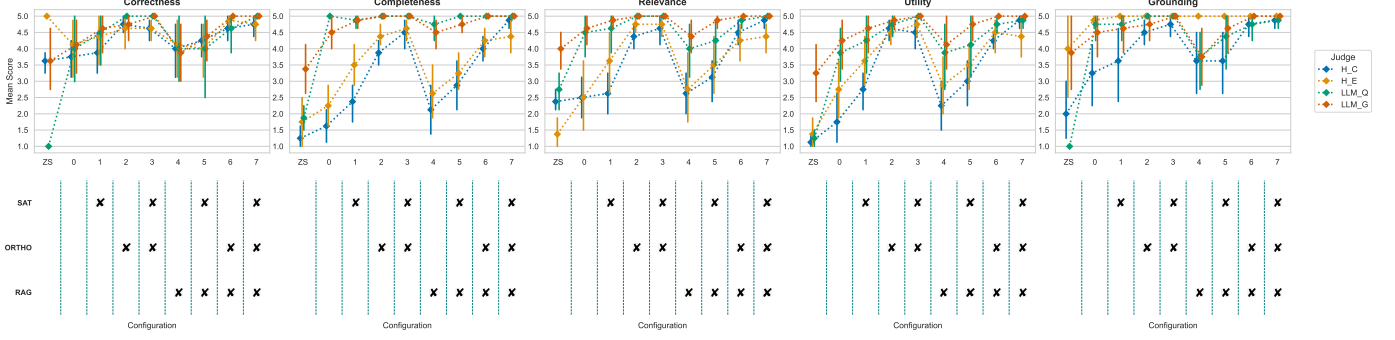


Figure 5: Preliminary evaluation results from the *Ablation Study* phase for the 4 evaluators and 5 metrics grouped by configuration. Differences can be observed based on active modules.

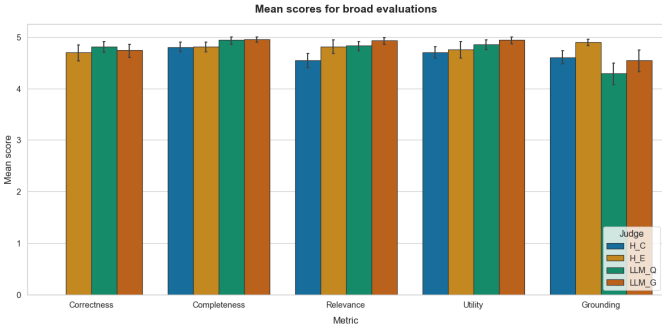


Figure 6: Average values for the 4 evaluators and 5 metrics in the *Broad* phase (complete system).

Shapley Values

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (v(S \cup \{i\}) - v(S)) \quad (1)$$

$$\begin{aligned} \phi_S(v) = & \frac{1}{3} [v(\{M_S\}) - v(\emptyset)] \\ & + \frac{1}{6} [v(\{M_S, M_O\}) - v(\{M_O\})] \\ & + \frac{1}{6} [v(\{M_S, M_R\}) - v(\{M_R\})] \\ & + \frac{1}{3} [v(\{M_S, M_O, M_R\}) - v(\{M_O, M_R\})] \end{aligned} \quad (2)$$

$$\begin{aligned} \phi_O(v) = & \frac{1}{3} [v(\{M_O\}) - v(\emptyset)] \\ & + \frac{1}{6} [v(\{M_S, M_O\}) - v(\{M_S\})] \\ & + \frac{1}{6} [v(\{M_O, M_R\}) - v(\{M_R\})] \\ & + \frac{1}{3} [v(\{M_S, M_O, M_R\}) - v(\{M_S, M_R\})] \end{aligned} \quad (3)$$

$$\begin{aligned} \phi_R(v) = & \frac{1}{3} [v(\{M_R\}) - v(\emptyset)] \\ & + \frac{1}{6} [v(\{M_S, M_R\}) - v(\{M_S\})] \\ & + \frac{1}{6} [v(\{M_O, M_R\}) - v(\{M_O\})] \\ & + \frac{1}{3} [v(\{M_S, M_O, M_R\}) - v(\{M_S, M_O\})] \end{aligned} \quad (4)$$

$$\begin{aligned} \phi_S(v) = & \frac{1}{3} [v(C_1) - v(C_0)] + \frac{1}{6} [v(C_3) - v(C_2)] \\ & + \frac{1}{6} [v(C_5) - v(C_4)] + \frac{1}{3} [v(C_7) - v(C_6)] \end{aligned} \quad (5)$$

$$\begin{aligned} \phi_R(v) = & \frac{1}{3} [v(C_2) - v(C_0)] + \frac{1}{6} [v(C_3) - v(C_1)] \\ & + \frac{1}{6} [v(C_6) - v(C_4)] + \frac{1}{3} [v(C_7) - v(C_5)] \end{aligned} \quad (6)$$

$$\begin{aligned} \phi_R(v) = & \frac{1}{3} [v(C_4) - v(C_0)] + \frac{1}{6} [v(C_5) - v(C_1)] \\ & + \frac{1}{6} [v(C_6) - v(C_2)] + \frac{1}{3} [v(C_7) - v(C_3)] \end{aligned} \quad (7)$$

4. Results and Discussion

5. Conclusions

Acknowledgements

Thanks to ...

Appendix A. Appendix A: Shapley values

References

Table A.5: Shapley

Judge	Metric	Module	Shapley Value	Relative Contribution (%)	Global Delta
H_C	Correctness	R	0.1667	16.67%	1.000
		O	0.7292	72.92%	
		S	0.1042	10.42%	
	Completeness	R	0.3958	12.18%	3.250
		O	2.0833	64.10%	
		S	0.7708	23.72%	
	Relevance	R	0.2292	9.65%	2.375
		O	1.8542	78.07%	
		S	0.2917	12.28%	
	Utility	R	0.2708	8.67%	3.125
		O	2.2083	70.67%	
		S	0.6458	20.67%	
	Grounding	R	0.2083	12.82%	1.625
		O	1.2083	74.36%	
		S	0.2083	12.82%	
H_E	Correctness	R	0.0208	3.33%	0.625
		O	0.5833	93.33%	
		S	0.0208	3.33%	
	Completeness	R	-0.0208	-0.98%	2.125
		O	1.5417	72.55%	
		S	0.6042	28.43%	
	Relevance	R	-0.1458	-7.78%	1.875
		O	1.4792	78.89%	
		S	0.5417	28.89%	
	Utility	R	-0.1042	-6.41%	1.625
		O	1.3333	82.05%	
		S	0.3958	24.36%	
	Grounding	R	0.0417	33.33%	0.125
		O	0.0417	33.33%	
		S	0.0417	33.33%	
LLM_Q	Correctness	R	-0.1458	-14.58%	1.000
		O	0.8542	85.42%	
		S	0.2917	29.17%	
	Completeness	R	-0.0625	0.00%	0.000
		O	0.0625	0.00%	
		S	0.0000	0.00%	
	Relevance	R	-0.2500	-50.00%	0.500
		O	0.6250	125.00%	
		S	0.1250	25.00%	
	Utility	R	-0.0625	-6.25%	1.000
		O	0.8125	81.25%	
		S	0.2500	25.00%	
	Grounding	R	-0.4792	-383.33%	0.125
		O	0.4583	366.67%	
		S	0.1458	116.67%	
LLM_G	Correctness	R	-0.0833	-9.52%	0.875
		O	0.6667	76.19%	
		S	0.2917	33.33%	
	Completeness	R	-0.0208	-4.17%	0.500
		O	0.3542	70.83%	
		S	0.1667	33.33%	
	Relevance	R	-0.0833	-22.22%	0.375
		O	0.2917	77.78%	
		S	0.1667	44.44%	
	Utility	R	0.0000	0.00%	0.750
		O	0.5000	66.67%	
		S	0.2500	33.33%	
	Grounding	R	-0.2083	-41.67%	0.500
		O	0.4792	95.83%	
		S	0.2292	45.83%	

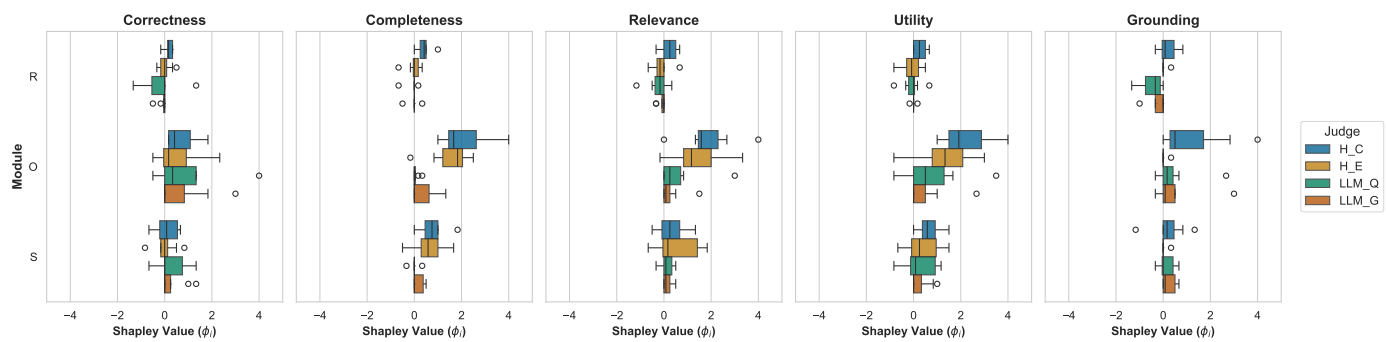


Figure 7: